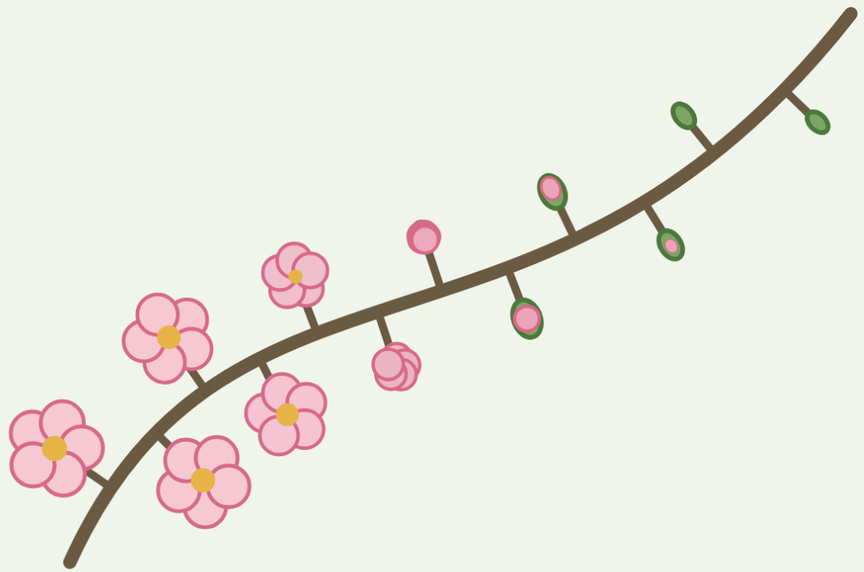
BIO144 · FS 2027 · WEEK 1

Introduction and a live data analysis

Data Analysis in Biology

Prof. Owen Petchey · University of Zurich



BIO144

MONDAY 22 FEBRUARY 2027 · 13:00–14:45

Today

13:00	Why data analysis, and what makes it good
13:15	How the course works, and the exam
13:35	How we will work together
13:40	AI assistants
13:50	RStudio Server
13:55	You collect some data
14:00	Short break
14:10	Live data analysis, from question to answer
14:40	What you should already know · this week	

Why should you care about data analysis?

THINK · PAIR · SHARE

Only you can answer that. Purpose comes from doing something **meaningful to you** and **useful to others**.

Show that a problem exists

Good analysis makes invisible problems visible, and helps others take them seriously.

Test solutions

Does an intervention work? For everyone? With unintended consequences?

Evaluate evidence that affects lives

Whether a drug works, or has serious side effects, shapes real decisions.

Question powerful claims

Marketing, policy and messaging often rely on selective statistics. What's missing? What's exaggerated?

Protect others

Especially those with less power: from children targeted by unhealthy products to communities facing environmental harm.

Serve something bigger

Science, public trust, health, biodiversity, fairness.

Questions you'll work on

Tap a card to see the question

Unit 1 · today
Reaction times
Your own data

Do male and female UZH students differ in reaction time?

Unit 2
Body fat
Exploring and graphing data in R

What does a dataset of body measurements look like, and how do the variables relate?

Units 3–4
Health spending
Countries, 2013

Is health-care spending in a country related to its child mortality? How strongly?

Unit 5
Blood pressure
A drug trial

Does the change in systolic blood pressure differ between treatment groups?

Unit 6
Primate milk · mayflies
Several explanatory variables

Is milk energy related to brain size, taking body mass into account? Do mayflies decline with farmland, taking altitude into account?

Unit 7
Intertidal algae
Herbivores on the shore

Does the effect of herbivory on algae depend on height in the intertidal?

Unit 8
Grouse ticks
Count data

How does the number of ticks on a chick change with altitude, taking years into account?

Unit 9
Pesticides
Binary data

How does the probability of death rise with dose, and does it differ among three products?

Unit 10
Plant communities
Many species at once

How does community composition change along a moisture gradient?

Unit 11
Sleep deprivation
Repeated measurements

How much does reaction time increase with each day of sleep deprivation?

Unit 12
Warming and nitrogen
Growth chambers

Does the effect of nitrogen on plant biomass depend on warming?

The exam
New case studies
You plan and analyse

Datasets you have not seen, with questions you answer by analysing them in R.

Every time, the same pattern



But how do we make sure the analysis is a good one?

HIGH-QUALITY DATA ANALYSIS

Four principles for reliable findings

PRINCIPLE 1

Define a focused question, then plan sampling and analysis to answer it

- Question *before* collecting or exploring data
- Match data collection to the question
- Plan (ideally register) the analysis

In BIO144: the first steps of every analysis, starting today.

PRINCIPLE 2

Use a model that accounts for the distribution and dependence of your data

- A distribution that suits the response (counts are not normal)
- Model all dependence (e.g. repeated measures)
- Check the assumptions

In BIO144: why we go from linear models to GLMs and mixed models.

PRINCIPLE 3

Emphasise effect sizes and biological relevance

- Report effect sizes with confidence intervals
- Discuss biological relevance
- De-emphasise p-values

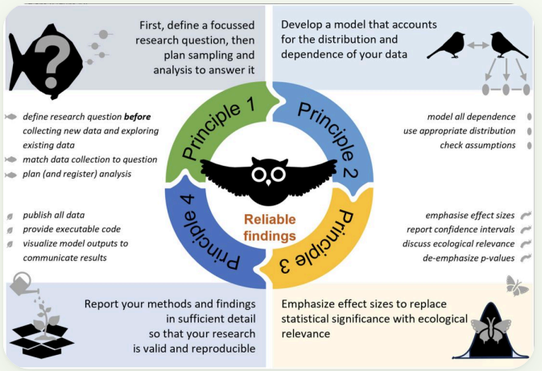
In BIO144: interpreting estimates and their uncertainty, every week.

PRINCIPLE 4

Report methods and findings so the work is valid and reproducible

- Publish all data
- Provide code others can run
- Visualise model outputs

In BIO144: every analysis is a script in R.



Popovic, G. et al. (2024). Four principles for improved statistical ecology. *Methods in Ecology and Evolution*, 15, 266–281.

The paper

HOW THE COURSE IS ORGANISED

Every week has four parts

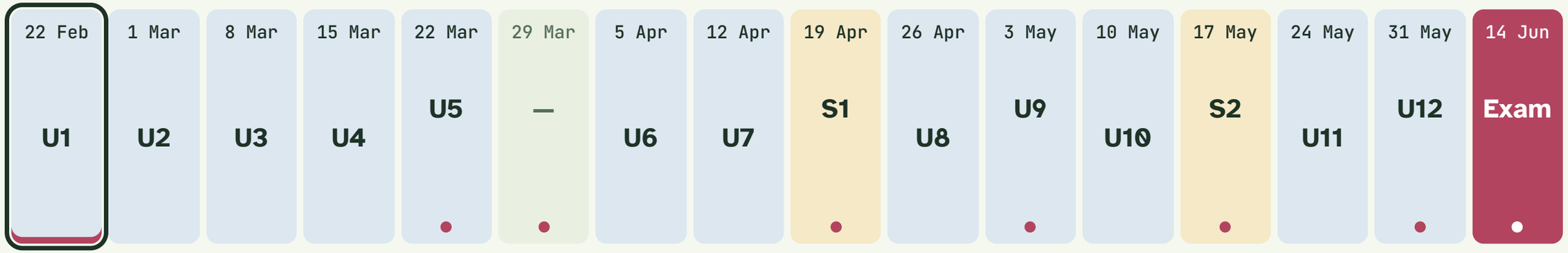
MON	TUE	WED	THU	FRI
Lecture 13:00–14:45	**Homework** before the practical	**Homework** when it suits you	**Practical A** 13:00–15:00	**Practical B** 13:00–15:00

Weekly practice quiz · any time, as often as you like — not marked

Your practical is on Thursday *or* Friday. Swapping for a week or two is fine without asking.

The semester at a glance

Tap a week



Unit (new content) · Self-study week, open practicals · Spring break · Final exam · Something to note

WEEK OF 22 FEB 2027

U1 · Introduction

Lecture Monday 13:00–14:45 · practicals Thursday and Friday 13:00–15:00.

THE CONCEPTUAL SIDE

Lectures, apart from today

- **No slides.**
- The **course book** contains what you need to know: one chapter per week.
- I talk, draw, and ask you questions. Little or no live coding: the code is in each chapter's walkthrough script.
- You don't need to prepare. If you'd like to:

  none · a little (skim the chapter) · more (read it, try the code)



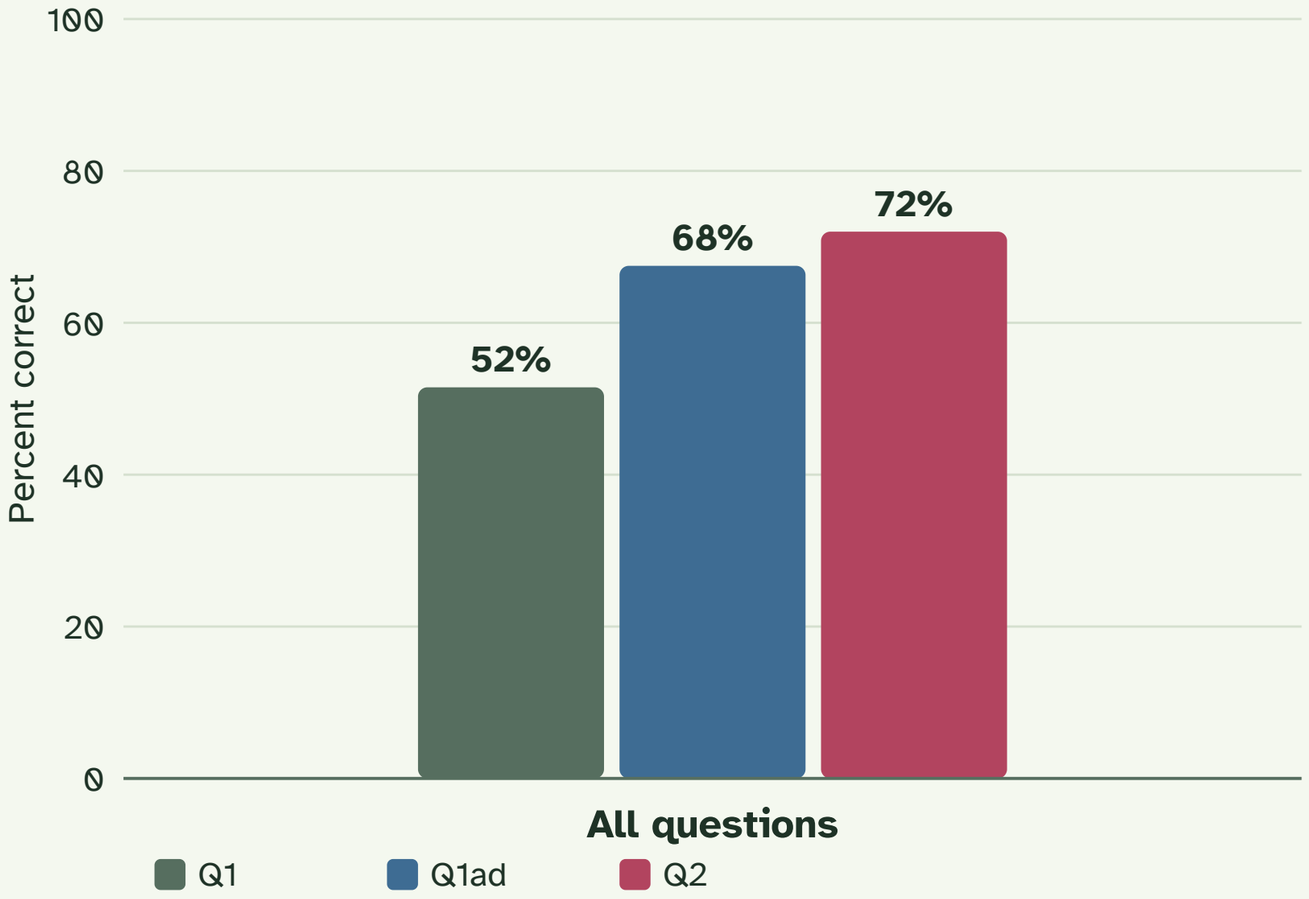
THINK · PAIR · SHARE

Talking to a neighbour helps, even when neither of you knew

Q1 answer alone · **Q1ad** same question after discussion · **Q2** a new, similar question, alone

Q1 **+ Q1ad** **+ Q2**

All questions **By difficulty**



Smith, M.K. et al. (2009). Why peer discussion improves student performance on in-class concept questions. *Science*, 323, 122–124. Values read from their Fig. 1 (approximate).

The team





Prof. Owen Petchey

Lecturer



Til Hämmig

Head teaching assistant; guest lectures in Units 8 and 9



Dr Frank Pennekamp

Examinations

TAs

Numerous teaching assistants

In every practical

Everything is on four websites

THE HUB

OLAT

Links to everything else, and the **discussion forum**: ask questions and make requests there.

HOW IT WORKS

Course information

General information, learning objectives, the schedule, and a page for each week.

opetchey.github.io/BIO144_Course_Information

WHAT TO KNOW

Course book

One chapter per week. Together with the learning objectives, it defines what you need for the exam.

opetchey.github.io/BIO144_Course_Book

WHAT TO DO

Homework, practicals, quizzes

One page per week, on the UZH RStudio Connect server. Off campus, connect to the **UZH VPN** first.

Linked from each week's page

Course information

Much of the material is new or newly revised. If something is unclear, too brief or wrong, please tell us.

Getting help

Practicals

Thursday and Friday: ask a TA, or your neighbour.

OLAT forum

Any time. We aim to answer within **two days**. Anonymous posts are possible.

Lectures

During and after.

Email

Fine, though I may ask you to post on the forum so everyone benefits.

Tough statistical questions: post them on the forum. Prof. Furrer has kindly agreed to help with those I can't answer.

The last day we can guarantee help with content: the last practical of the semester (Fri 4 June).

FROM THE 2026 COURSE EVALUATION

You told us, so in 2027...

YOU TOLD US (2026)	WHAT WE DID FOR 2027
Interactive lectures, examples and the forum were appreciated. Practicals, quizzes, TAs and the course book help a lot.	**We kept all of that.**
Live coding in lectures was too fast to follow or copy, and the code wasn't available afterwards.	**Little or no live coding in lectures. Every chapter has a code walkthrough script, with a “Practical toolbox”, to work through at your own pace.**
Please give us solutions, or a way to check our work.	**An example solution script for every practical (last tab). All quiz answers re-checked.**
R is hard for beginners, and Week 1 was too hard.	**Unit 1 practical rebuilt: log in, make a project, import your data, fix a script. “Getting R & RStudio” in the self-tests.**
Three websites were confusing, and it wasn't clear how much of the book mattered.	**Every Unit page has the same layout and its learning objectives: those plus the chapter define the exam. One page lists all homework, practicals and quizzes.**
Holidays meant the Friday group lost practicals.	**Every holiday swap is in the schedule. Two self-study weeks with open practical sessions.**
“Prof. Petchey talks for hours on end.”	**Sit back, open popcorn, enjoy :)**

The final examination

Friday

18 June 2027

10:00–12:00 · Irchel

Resit: Thursday 26 August 2027, 14:15–16:15
Mandatory Probeprüfung in mid to late May

1 **Case studies:** you analyse datasets in R and answer questions about them.

A4 **Three sheets** of A4 notes, both sides. One translation dictionary.

⌀ **No AI** of any kind, including Copilot. Prohibited materials and tools are strictly controlled.

40 About 40 questions: single choice, multiple choice or **numeric** (default 2 decimal places).

⌨ Your laptop and **RStudio Server**. How the exam setup works will be announced by late May.

✓ A **mock exam** on OLAT, and the weekly quizzes, give you practice.

Best preparation: do analyses yourself, every week. Be quick at importing data.



THINK · PAIR · SHARE

How hard will this course be?

4 credit points ≈ 120 hours

⅓ in class | **⅔ self-study**

Weeks are not all the same amount of work: look ahead and plan early.

Keep up, and reach out early if you feel you're falling behind.

MY ROLE, YOUR ROLE, AND HOW WE'LL WORK TOGETHER

Becoming great data analysts

- Everyone here can learn to think analytically and statistically.
- I'll hold you to **high standards**, because I believe you can meet them. My job is to support you in reaching them.
- **Struggle is part of growth**, not a sign of failure.
- Stress before a challenge is normal: a faster heart and breathing are your body giving you more resources.



When did someone believe in you more than you believed in yourself? What happened?

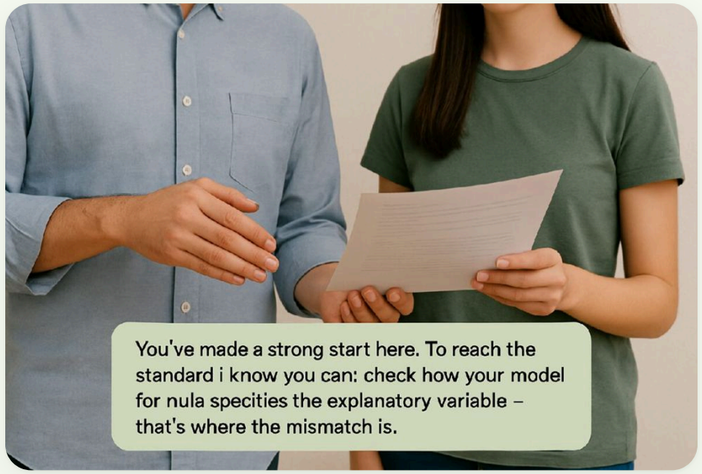
High standards and wise feedback

> **“I'm giving you these comments because I have high standards and I know that you can meet them.”**

“Wise feedback”: Cohen, G.L., Steele, C.M. & Ross, L.D. (1999). *Personality and Social Psychology Bulletin*, 25, 1302–1318.

- Feedback will be honest and specific.
- It comes from belief in your potential, not criticism of you as a person.
- If it feels like criticism, try to reframe it as being challenged to grow.

This class is a safe space for you, and a challenging space for you.



Klicker

THINK · PAIR · SHARE

How comfortable are you receiving critical feedback?



I may be a professor, but I'm also human

Call me **Professor Petchey** or **Owen**, whatever you're comfortable with.

(Probably not “dude” or “mate”.)

We are here to support you: to give you the knowledge of how to learn more, and the confidence that you can.

AI ASSISTANTS

Helpful, or harmful to your learning?

Vote, then tap

“What is the mode of a distribution?”

Good use: help understanding

“I didn't understand that explanation. Can you give an easier one, and show it graphically?”

Good use: help understanding

“Here is my code and the error. What does the error mean?”

Often good, after you've tried yourself. You can't do this in the exam.

“Write R code to import my data and give the number of missing values in each variable.”

Take care: can save time, but read and understand every line

“Which statistical test should I use for these data?”

Take care: often subtly wrong (e.g. the wrong model for binary data). Use your own judgement.

“Import the data, fit a regression, check diagnostics, summarise, make a nice figure, put it in a Quarto file. Do it while I go and chill.”

You learn nothing, and you can't check it's right

Your relationship with AI assistants

A strong relationship

The AI is a knowledgeable collaborator that complements your skills and supports your growth. **You** provide context, direction and critical judgement.

A poor relationship

You use output you don't understand, copying and pasting. Errors slip through, and your own skills stagnate.

The knowledge *you* bring is what makes the relationship good. So: **minimise your use of AI during the course** (and of the example solutions). Do the work, get the benefit.

In the exam, prohibited tools are strictly controlled: **no generative AI at all**, and no Copilot in RStudio. Practise without it.

THINK · PAIR · SHARE

What can I use an AI assistant for in BIO144?

And what should I not use it for?

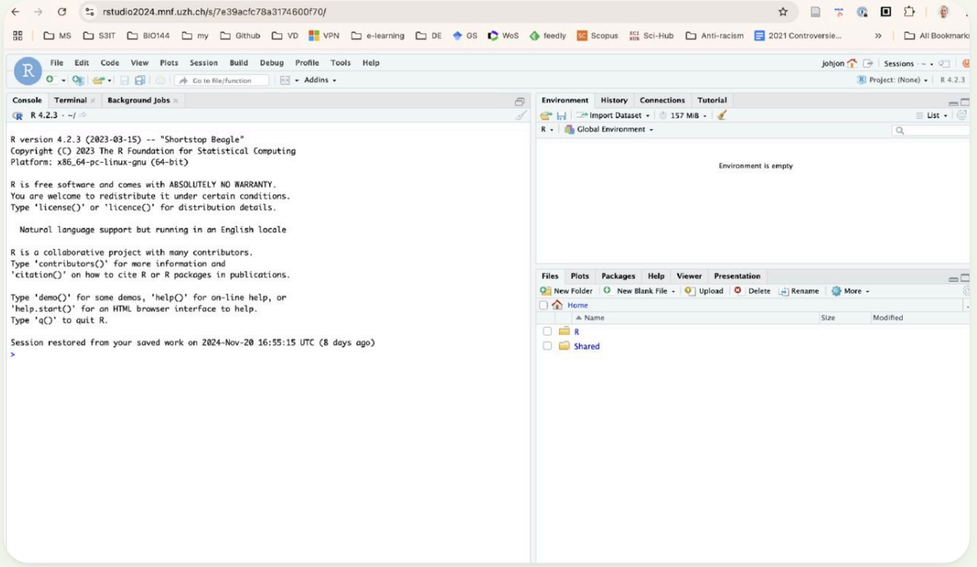
Klicker

RStudio Server

RStudio in your web browser: rstudio.mnf.uzh.ch, with your UZH login.

- **You must use it in the exam**, so make sure you can log in and work there.
- Good: most of the setup is done for you.
- Less good: you need an internet connection.
- Everyone in the OLAT course is registered. Not in OLAT? No access.
- Prefer your own laptop? Fine: see “Getting R & RStudio” in the Previous Knowledge self-tests.

The Unit 1 practical starts by logging in.



rstudio.mnf.uzh.ch

WE NEED SOME DATA

Your turn: measure your reaction time

1. Open the **reaction time test** (humanbenchmark.com).
2. Do it **five times**. Note each time, in **milliseconds**.
3. Enter your five times, and answer the other questions, in the **form**.

Reaction time test

Data form

It's anonymous. After the lecture the class data are published as `reaction_times_2027.csv`, for the practical.

Both links are on the course information site, Unit 1 page, “Lecture”.

Short break

10:00

Start | −1 min | +1 min | Reset

LIVE DATA ANALYSIS

From question to answer in under an hour

- It's a demonstration: a feel for what's involved.
- You'll understand some of it, and not other parts. That's expected.
- **Keep notes about what you don't understand.** Look for the answers as the course goes on.

The workflow in twelve steps

1. Question
2. Expectation
3. Planned graph & analysis
4. Selection of subjects
5. How data are collected
6. Ethics, permissions
7. Data collection
8. Data wrangling
9. Visualisation
10. Statistical test
11. Critical thinking
12. Report, communicate

The first six steps all happen **before** any data are collected. That's Principle 1 in action.

1 **Question**
2 Expectation
3 Planned graph & analysis
4 Selection of subjects
5 How data are collected
6 Ethics, permissions

before data ↑

7 Data collection
8 Data wrangling
9 Visualisation
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What should our question be?

- Of interest to us, and of some importance.
- Answerable with data we can collect, within our constraints.

Do the reaction times of male and female students at the University of Zurich differ?

“Male” and “female”: sex assigned at birth. Why care? Reaction times matter for safety (driving) and in sport (the start of a sprint).

1 Question
2 **Expectation**
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Expectation

There has been quite a lot of research already.

In general, males tend to have faster reaction times than females. So we expect the same for students at UZH.

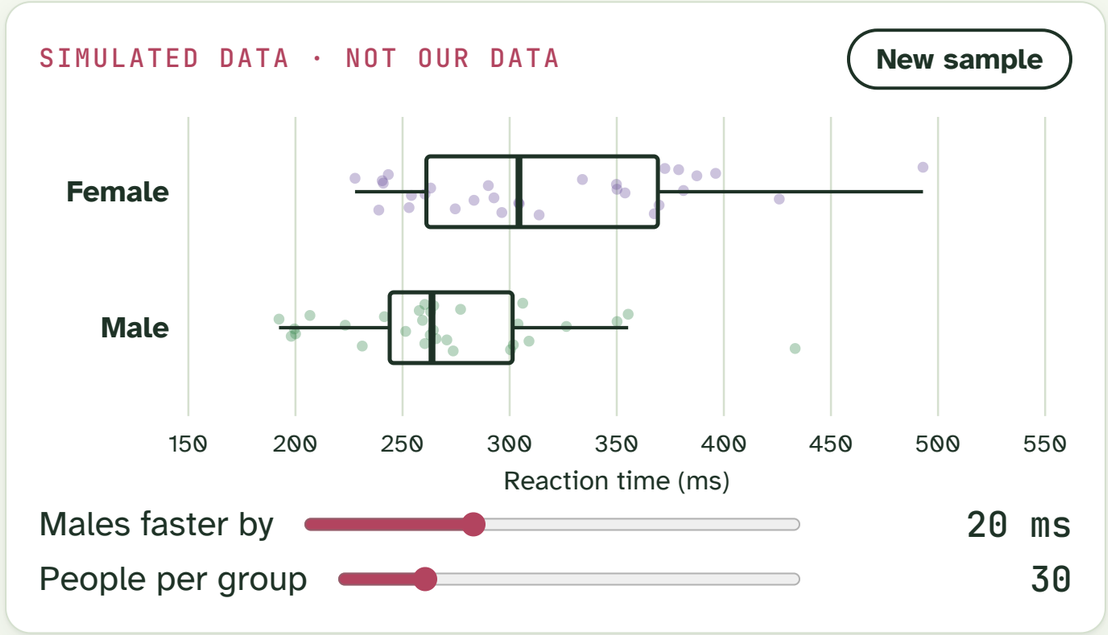
Curious? Lipps et al. (2011) looked at reaction times of sprinters at the Beijing Olympics (PLOS ONE).

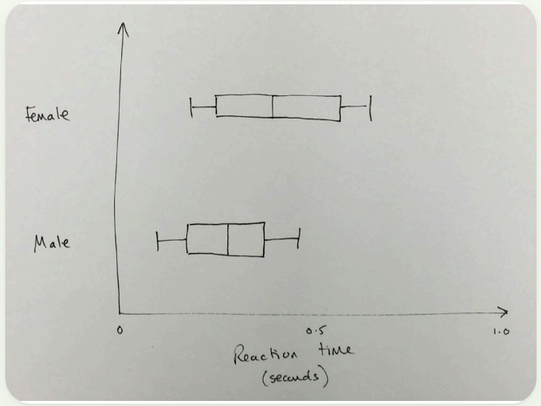
1 Question
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4 Selection of subjects
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What graph will answer the question?





Sketch the graph **before** collecting data: box plots of reaction time (ms) for females and males.

1 Question
2 Expectation
3 **Planned graph & analysis**
4 Selection of subjects
5 How data are collected
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What statistical test? What assumptions?

Response: reaction time (continuous). Explanatory variable: sex (two groups).

A two-sample t-test compares the means of two groups.

A natural first choice here. In later weeks you'll see it is a special case of a linear model.

Its assumptions? (tap)

Independent observations · values roughly normally distributed in each group · similar variances in the two groups.

1 Question
2 Expectation
3 Planned graph & analysis
4 **Selection of subjects**
5 How data are collected
6 Ethics, permissions
before data ↑
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Selection of subjects

Ideally a random sample of the population we want to draw conclusions about.

Are you a random sample of… (tap)

all humans?

No: an age group, a country, university students.

all UZH students?

Closer, but only biology students in one course, who chose to take part.

students in this lecture?

Those who came today and filled in the form.

How subjects are selected limits who our conclusions apply to.

1 Question
2 Expectation
3 Planned graph & analysis
4 Selection of subjects
5 **How data are collected**
6 **Ethics, permissions**

before data ↑

7 **Data collection**
8 Data wrangling
9 Visualisation
10 Statistical test
11 Critical thinking
12 Report, communicate

Collecting data, and ethics

How? An online test, and a form. **Done: you just did it.**

Because these are data about people, we should think about ethics:

Consent

Taking part was voluntary.

Anonymity

No names. Could anyone be identified?

Use and sharing

For teaching, published for the practical.

1 Question
2 Expectation
3 Planned graph & analysis
4 Selection of subjects
5 How data are collected
6 Ethics, permissions

before data ↑

7 Data collection
8 **Data wrangling**
9 **Visualisation**
10 **Statistical test**
11 Critical thinking
12 Report, communicate

Data wrangling, visualisation, test

Live in RStudio →

Why a script in R, rather than pointing and clicking? It makes the analysis:

Efficient **Consistent** **Repeatable** **Reliable** **Readable**
Robust **Persistent** **Sharable** **Scalable**

That's Principle 4 in action.

1 Question
2 Expectation
3 Planned graph & analysis
4 Selection of subjects
5 How data are collected
6 Ethics, permissions

before data ↑

7 Data collection
8 Data wrangling
9 Visualisation
10 Statistical test
11 **Critical thinking**
12 **Report, communicate**

Critical thinking, and reporting

How big is the difference?

Is it biologically relevant? (Principle 3)

How precise?

The confidence interval of the difference.

Other explanations?

Could something other than sex explain it?

To whom does it apply?

Given how the subjects were selected.

Report: a clear figure, and a sentence with the estimate, its uncertainty, and the test.

The four principles, in today's analysis

PRINCIPLE 1

Question first

Question, expectation, graph and test, all decided before collecting data.

PRINCIPLE 2

A suitable model

The t-test's assumptions. Later: counts, binary data, repeated measures.

PRINCIPLE 3

Effect sizes

How big is the difference, how precise, does it matter?

PRINCIPLE 4

Reproducible

A script anyone can re-run, on data that are shared.

ACTIVATE YOUR PRIOR LEARNING

What it helps to know already

0 of 31 feel familiar

SUMMARISING DATA

- mean, median, mode
- range, quartiles
- standard deviation
- standard error
- skew
- kurtosis
- uni-, bi-, multimodal

TYPES OF VARIABLE

- numeric
- continuous
- discrete
- categorical
- response
- explanatory

STATISTICAL IDEAS

- expected value
- parameter and estimate
- population vs sample
- central limit theorem
- hypothesis testing
- degrees of freedom
- p-value

PROBABILITY

- 5 heads in 7 tosses?
- normal
- binomial
- Poisson
- exponential
- histograms

METHODS AND GRAPHS

- t-test
- linear regression
- equation of a line
- box plots
- scatterplots

NOTATION

$$\bar{x} = (1/n) \Sigma x_i$$

$$s^2 = 1/(n-1) \Sigma (x_i - \bar{x})^2$$

More in Chapter 1 of the course book

Previous Knowledge self-tests (UZH network or VPN). Not sure about some? That's what they are for.

This week

BEFORE THE PRACTICAL

Homework

Read Chapter 1 of the course book. Work through the Unit 1 homework.

THU 25 OR FRI 26 FEB

Practical

Log in to RStudio Server, make a project, import *your* data, fix a broken script.

ANY TIME

Weekly quiz

Check your understanding. Not marked.

IF NEEDED

Previous knowledge

The self-tests, and help with installing R and RStudio.

Unit 1 page

See you in the practical. Have fun!

Questions any time: the OLAT forum.